MATHS MANTHAN
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CBSE CLASS 10 | MATHEMATICS

AREAS RELATED TO CIRCLES

Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

See the region. Dissect the geometry.
Subtract with absolute precision.

Sectors | Segments | Annulus | Combinations | Board Traps

A student-first Maths Manthan conceptual and visual revision resource



What This Handbook Includes

- Exact proofs of sector arc length and area formulas
- Universal $\triangle OAB$ trigonometric area derivation
- Inscribed/circumscribed polygons and combination figures
- Annulus, athletic track, and tethered grazing dynamics
- Clock face sweeps, automobile wipers, and brooches
- Master formula matrices and board audit protocols

Pedagogical Strategy: Foundations First

Never blindly memorize formulas. Every sector is simply an angular fraction $\left(\frac{\theta}{360^\circ}\right)$ of a complete circle. Every segment is an elementary difference of two known shapes:

$$\text{Segment} = \text{Sector } \mathbf{OAB} - \triangle\mathbf{OAB}$$

Always factor out common algebraic terms (πr^2, r^2) before evaluating numbers!

CBSE 2026–27 Curriculum Verification & Rationalized Scope

Official Scope (Prescribed & Tested): Areas of sectors and segments of a circle (minor and major); length of corresponding arc; perimeter of a sector; area of combinations of plane figures involving circles, quadrants, semicircles, triangles, rectangles, and regular polygons. Central angles $\theta \in \{30^\circ, 45^\circ, 60^\circ, 90^\circ, 120^\circ\}$ are strictly in scope.

Excluded / Non-Examinable: Calculus/integration area derivations; frustum of a cone cross-sections (shifted to Ch 12); geometric proofs of triangle similarity area ratios (deleted from Ch 6).

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Prepared for learning and conceptual mastery – Strictly aligned with CBSE Class 10 Syllabus 2026–27.

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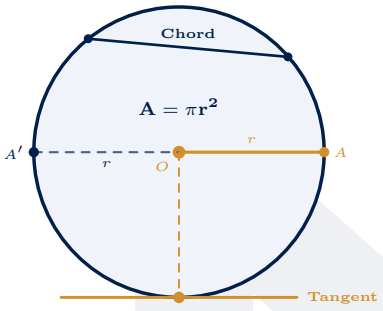
CORE SYLLABUS

AREAS RELATED TO CIRCLES

Module 1 | Foundations: Anatomy, The Geometry of π & Wheel Kinematics

1.1 Anatomy of a Circle and Prerequisite Metrics

A **circle** is the locus of all points in a plane situated at a fixed distance (radius r) from a fixed center O.



Fundamental Metrics of a Circle of Radius r:

- **Diameter (d):** $\mathbf{d = 2r \iff r = d/2}$
- **Circumference (C):** Perimeter of boundary:

$$\mathbf{C = 2\pi r = \pi d}$$

- **Enclosed Area (A):** 2D region bounded by curve:

$$\mathbf{A = \pi r^2 = \pi (d/2)^2 = \frac{1}{4}\pi d^2}$$

- **Semicircle ($180°$):**

$$\text{Area} = \mathbf{\frac{1}{2}\pi r^2}, \quad \text{Perimeter} = \mathbf{\pi r + 2r}$$

- **Quadrant ($90°$):**

$$\text{Area} = \mathbf{\frac{1}{4}\pi r^2}, \quad \text{Perimeter} = \mathbf{\frac{1}{2}\pi r + 2r}$$

1.2 The Geometric Constant π (Pi): Concept and Conventions

By definition, π is the ratio of the circumference of any circle to its diameter: $\pi = \frac{C}{2r}$. π is an **irrational number** ($\pi \approx 3.14159265\ldots$). Aryabhata gave the approximation $\pi \approx \frac{62832}{20000} = 3.1416$.

- **CBSE Rule:** Always use $\pi = \mathbf{\frac{22}{7}}$ unless explicitly instructed to take $\pi = \mathbf{3.14}$.

1.3 Linear Kinematics of a Rolling Wheel (Revolutions & Distance)

When a circular wheel of radius r completes **one revolution** without slipping, the linear distance covered along the ground is exactly equal to its circumference:

$$\text{Distance in 1 revolution} = \mathbf{2\pi r} \implies \text{Total Distance: } \mathbf{D = N \times 2\pi r} \iff \mathbf{N = \frac{D}{2\pi r} = \frac{D}{\pi d}}$$

Worked Archetype 1 | Rotational Speed & Number of Revolutions

Problem: The wheels of a car are of diameter 80 cm each. How many complete revolutions does each wheel make in 10 minutes when the car travels at 66 km/h? (Take $\pi = \frac{22}{7}$)

Step 1: Unify units: $d = 80$ cm $\implies r = 40$ cm $= 0.4$ m. Speed $v = \frac{66 \times 1000 \text{ m}}{60 \text{ min}} = 1100$ m/min.

Step 2: Total distance in 10 minutes: $D = 1100$ m/min $\times$ 10 min $= 11{,}000$ m $= 1{,}100{,}000$ cm.

Step 3: Distance in 1 revolution: $C = 2\pi r = 2 \times \frac{22}{7} \times 40 = \frac{1760}{7}$ cm.

Step 4: Revolutions: $N = \frac{D}{C} = \frac{1{,}100{,}000 \times 7}{1760} = \mathbf{4375}$ complete revolutions.

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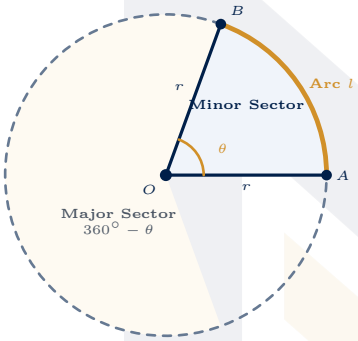
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Module 2 | Sectors: Arc Length, Sector Area & Perimeter

2.1 The Concept of a Circular Sector

A **sector** is the portion of a circular disc enclosed by two radii (OA and OB) and the arc (AB) subtending a central angle θ at the center O.

- **Minor Sector:** The sector subtending central angle $\theta < 180°$.
- **Major Sector:** The remainder of the circular disc, subtending central angle $(360° - \theta)$.



Universal Sector Formulas (Angle θ in degrees):

- **Length of Arc (l):**

$$\mathbf{l} = \frac{\theta}{\mathbf{360°}} \times \mathbf{2\pi r} = \frac{\theta}{\mathbf{180°}} \times \mathbf{\pi r}$$

- **Area of Sector (A):**

$$\mathbf{A} = \frac{\theta}{\mathbf{360°}} \times \mathbf{\pi r^2}$$

- **Relation Between Arc Length l and Area A:**

$$\mathbf{A} = \frac{\mathbf{1}}{\mathbf{2}} \cdot \mathbf{l} \cdot \mathbf{r} \quad \left(\text{Like } \frac{1}{2} \times \text{base} \times \text{height}\right)$$

- **Perimeter of Sector (P):**

$$\mathbf{P} = \mathbf{l} + \mathbf{2r} = \frac{\theta}{\mathbf{360°}} \times \mathbf{2\pi r} + \mathbf{2r}$$

- **Area of Major Sector:**

$$\mathbf{A}_{\text{major}} = \mathbf{\pi r^2} - \mathbf{A}_{\text{minor}} = \frac{\mathbf{360° - \theta}}{\mathbf{360°}} \times \mathbf{\pi r^2}$$

2.2 The Fatal "Sector Perimeter" Board Trap

TRAP ALERT

Arc Length $\neq$ Perimeter of Sector! In board exams, many students write $P = l$. An arc is an **open curve**. To form a closed sector, you **must add both bounding radii** OA and OB:

$$\text{Perimeter of Sector} = \text{Arc Length } (\mathbf{l}) + \mathbf{2} \times \text{Radius } (\mathbf{r}) = \mathbf{l + 2r}$$

Worked Archetype 2 | Arc Length, Sector Area, and Perimeter

Problem: In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find: (i) length of the arc, (ii) area of the sector formed by the arc, and (iii) perimeter of the sector. (Take $\pi = \frac{22}{7}$)

Solution: Given $r = 21$ cm, central angle $\theta = 60°$.

1. **Length of the arc (l):** $l = \frac{60°}{360°} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 2 \times 22 \times 3 = \mathbf{22}$ cm.
2. **Area of the sector (A):** $A = \frac{60°}{360°} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{6} \times 22 \times 3 \times 21 = 11 \times 21 = \mathbf{231}$ cm^2.
   Alternative Verification: $A = \frac{1}{2}lr = \frac{1}{2} \times 22 \times 21 = \mathbf{231}$ cm^2.
3. **Perimeter of the sector (P):** $P = l + 2r = 22 + 2(21) = 22 + 42 = \mathbf{64}$ cm.

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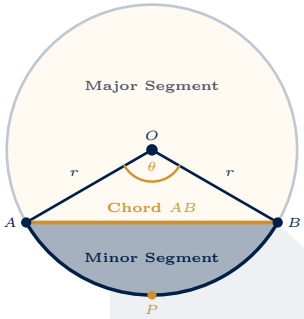
AREAS RELATED TO CIRCLES

Module 3 | Segments of a Circle: Dissection, Area & Formulas

3.1 The Geometric Definition of a Circular Segment

A **segment** of a circle is the region bounded by a chord (AB) and the corresponding arc connecting its endpoints:

- **Minor Segment:** Region bounded by chord AB and minor arc APB.
- **Major Segment:** Region bounded by chord AB and major arc AQB.



The Universal Segment Dissection Axiom:

$$\mathbf{A}_{\text{minor}} = \text{Area(Sector } \mathbf{OAPB}) - \text{Area}(\triangle\mathbf{OAB})$$

$$\mathbf{A}_{\text{major}} = \text{Area of Circle} - \mathbf{A}_{\text{minor}} = \pi\mathbf{r^2} - \mathbf{A}_{\text{minor}}$$

Universal Triangle Formula:

$$\text{Area}(\triangle\mathbf{OAB}) = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{r^2}\sin\theta$$

$$\mathbf{A}_{\text{minor}} = \mathbf{r^2}\left[\frac{\pi\theta}{\mathbf{360^\circ}} - \frac{\mathbf{1}}{\mathbf{2}}\sin\theta\right]$$

3.2 Rigorous Derivation: Area$(\triangle OAB) = \frac{1}{2}r^2\sin\theta$

Draw $OM \perp AB$. In isosceles triangle $\triangle OAB$ ($OA = OB = r$):

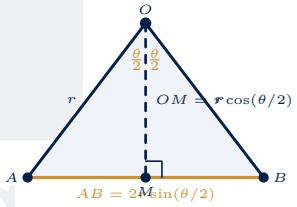
1. Altitude OM bisects chord AB and vertex angle $\angle AOB = \theta$:

$$\angle AOM = \angle BOM = \frac{\theta}{2}$$

2. In right $\triangle OAM$:

$$AM = r\sin(\theta/2) \implies AB = 2r\sin(\theta/2)$$

$$OM = r\cos(\theta/2)$$



3. Triangle Area:

$$\text{Area} = \frac{1}{2}\cdot AB\cdot OM = \frac{1}{2}[2r\sin(\theta/2)][r\cos(\theta/2)]$$

$$\text{Area} = r^2\sin(\theta/2)\cos(\theta/2) = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{r^2}\sin\theta$$

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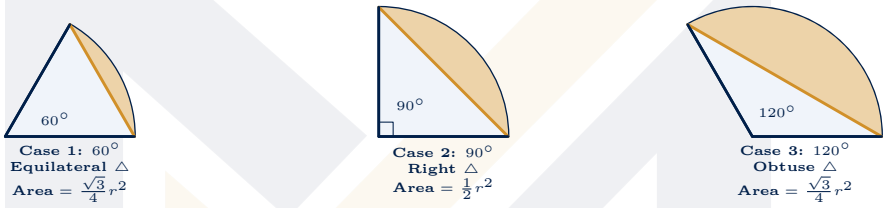
Module 4 | Master Angle Evaluation: $\theta \in \{60^\circ, 90^\circ, 120^\circ\}$

4.1 Exact Evaluation of $\triangle OAB$ Across Board-Mandated Angles

The CBSE Class 10 examination focuses on three primary central angles for segment problems:

θ	$\sin\theta$	$\frac{\theta}{360^\circ}$	$\triangle OAB$ Type	**Area**$(\triangle OAB)$ **Formula**	Visual Geometric Shortcut
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{6}$	Equilateral	$\frac{\sqrt{3}}{4}\mathbf{r^2}$	Since $OA = OB$ and $\angle O = 60^\circ$, all angles are 60°.
90°	1	$\frac{1}{4}$	Right-Isosceles	$\frac{1}{2}\mathbf{r^2}$	Base $= r$, Perpendicular Altitude $= r \implies \frac{1}{2}r \cdot r$.
120°	$\frac{\sqrt{3}}{2}$	$\frac{1}{3}$	Obtuse Isosceles	$\frac{\sqrt{3}}{4}\mathbf{r^2}$	Base $= r\sqrt{3}$, Altitude $= \frac{1}{2}r \implies \frac{1}{2}(r\sqrt{3})(\frac{1}{2}r)$.

4.2 Geometric Visual Architectures for 60°, 90°, and 120°



Worked Archetype 3 | Minor Segment with $\theta = 120^\circ$

Problem: A chord of a circle of radius 12 cm subtends an angle of 120° at the centre. Find the area of the corresponding minor segment. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$)

Solution: Given $r = 12$ cm, $\theta = 120^\circ$.

Step 1: Area of Sector OAB: $A_{\text{sector}} = \frac{120^\circ}{360^\circ} \times 3.14 \times 12^2 = \frac{1}{3} \times 3.14 \times 144 = 3.14 \times 48 = \mathbf{150.72}$ cm^2.

Step 2: Area of $\triangle OAB$: $\text{Area}(\triangle OAB) = \frac{1}{2}r^2 \sin 120^\circ = \frac{1}{2}(144)\left(\frac{\sqrt{3}}{2}\right) = 36\sqrt{3} = 36 \times 1.73 = \mathbf{62.28}$ cm^2.

Step 3: Area of Minor Segment: $\mathbf{A}_{\text{minor}} = A_{\text{sector}} - \text{Area}(\triangle OAB) = 150.72 - 62.28 = \mathbf{88.44}$ cm^2.

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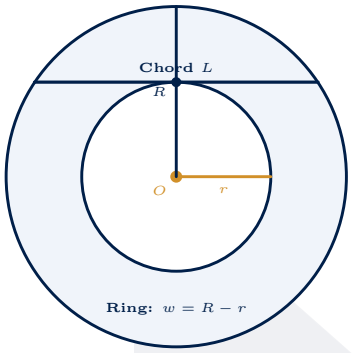
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Module 5 | Annulus (Circular Ring) & Athletic Track Geometry

5.1 Geometry of the Circular Ring (Annulus)

An **annulus** is the region enclosed between two concentric circles with radii r and R ($R > r$):



Master Formulas for Annulus (Outer R, Inner r):

- **Width of Ring (w): $\mathbf{w = R - r}$**
- **Area of Ring (A):**

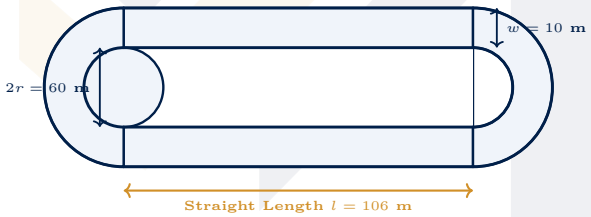
$$\mathbf{A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2) = \pi(R + r)(R - r)}$$

- **In Terms of Ring Width w: $\mathbf{A = \pi w(2r + w)}$**
- **Tangent Chord Theorem:** If a chord of outer circle of length L touches inner circle:

$$\mathbf{A = \frac{\pi L^2}{4}} \quad \left(\text{since } R^2 - r^2 = (L/2)^2\right)$$

5.2 The Athletic Running Track Model (Board Classic)

A standard running track has two parallel straight lines joined by two semicircular ends:



Running Track Equations (Straight length l, inner radius r, outer radius $R = r + w$):

- **Inner Perimeter: $\mathbf{P_{inner} = 2l + 2\pi r}$**
- **Outer Perimeter: $\mathbf{P_{outer} = 2l + 2\pi R = 2l + 2\pi(r + w)}$**
- **Area of Running Track:**

$$\text{Area} = \mathbf{2(l \times w) + \pi(R^2 - r^2) = 2lw + \pi w(2r + w)}$$

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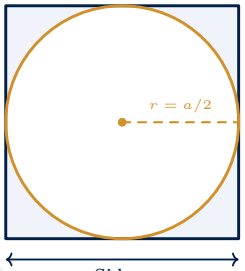
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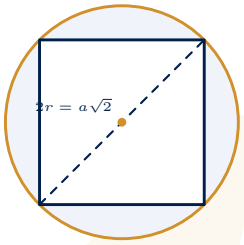
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Module 6 | Inscribed & Circumscribed Circles with Polygons

6.1 Circle Inscribed in a Square vs Square Inscribed in a Circle



Inscribed Circle ($r = a/2$)
Shaded $= a^2 - \pi(a/2)^2 = \left(1 - \frac{\pi}{4}\right)a^2 \approx \frac{3}{14}\mathbf{a^2}$



Circumscribed Circle ($a = r\sqrt{2}$)
Shaded $= \pi r^2 - 2r^2 = (\pi - 2)r^2 \approx \frac{8}{7}\mathbf{r^2}$

Comparative Ratio Summary for Squares and Circles:

- **Inscribed Circle:** $\frac{\text{Area of Circle}}{\text{Area of Square}} = \frac{\pi}{4} \approx \frac{11}{14} \approx 78.5\% \implies$ Shaded Corners $\approx \frac{3}{14}\mathbf{a^2} \approx 21.5\%$.
- **Circumscribed Circle:** $\frac{\text{Area of Square}}{\text{Area of Circle}} = \frac{2}{\pi} \approx \frac{7}{11} \approx 63.6\% \implies$ Shaded Boundary $\approx \frac{4}{11}\pi\mathbf{r^2}$.

6.2 Circles and Equilateral Triangles (Inradius & Circumradius)

For an equilateral triangle of side a and height $h = \frac{\sqrt{3}}{2}a$:

- **Inradius (r):** $\mathbf{r = \frac{h}{3} = \frac{a}{2\sqrt{3}}} \implies$ Area of Inscribed Circle $= \pi\mathbf{r^2} = \frac{\pi\mathbf{a^2}}{\mathbf{12}}$.
- **Circumradius (R):** $\mathbf{R = \frac{2h}{3} = \frac{a}{\sqrt{3}} = 2r} \implies$ Area of Circumscribed Circle $= \pi\mathbf{R^2} = \frac{\pi\mathbf{a^2}}{\mathbf{3}} = \mathbf{4} \times$ (Incircle Area).
- **Area of Equilateral Triangle:** Area $= \frac{\sqrt{3}}{4}a^2$.

Worked Archetype 4 | Equilateral Triangle & Circumscribed Circle

Problem: A circular table cover of radius 32 cm has an equilateral triangle ABC inscribed inside it. Find the area of the design pattern outside the triangle.

Step 1: Circle Area: $R = 32$ cm $\implies A_{\text{circle}} = \frac{22}{7} \times 32 \times 32 = \mathbf{\frac{22528}{7}}$ cm^2.

Step 2: Triangle Side a: $R = \frac{a}{\sqrt{3}} \implies a = 32\sqrt{3}$ cm.

Step 3: Triangle Area: Area $= \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}(32\sqrt{3})^2 = \frac{\sqrt{3}}{4}(1024 \times 3) = \mathbf{768\sqrt{3}}$ cm^2.

Step 4: Design Area: Design Area $= \left(\mathbf{\frac{22528}{7} - 768\sqrt{3}}\right)$ cm^2.

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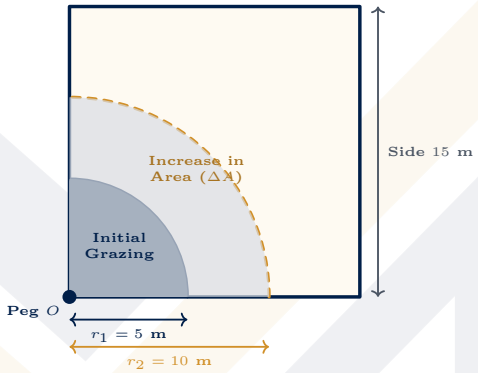
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Module 7 | The Tethered / Grazing Animal Dynamics

7.1 Physical Intuition: Corner Angle as Sector Angle

When an animal is tethered by a rope of length r to a peg at a corner of a field:

1. The length of the rope acts as the **radius**: $\mathbf{r}$ = rope length.
2. The interior corner angle of the field acts as the **central angle** θ:
   - Corner of a **Square / Rectangle**: $\theta = \mathbf{90^\circ} \implies$ Grazing Area $= \frac{1}{4}\pi\mathbf{r^2}$ (Quadrant).
   - Corner of an **Equilateral Triangle**: $\theta = \mathbf{60^\circ} \implies$ Grazing Area $= \frac{1}{6}\pi\mathbf{r^2}$.
   - Pegged at **middle of straight wall**: $\theta = \mathbf{180^\circ} \implies$ Grazing Area $= \frac{1}{2}\pi\mathbf{r^2}$ (Semicircle).
   - Open field (no walls): $\theta = \mathbf{360^\circ} \implies$ Grazing Area $= \pi\mathbf{r^2}$ (Full Circle).



Increase in Grazing Area When Rope is Extended from r_1 to r_2:

$$\mathbf{\Delta A} = \frac{\theta}{\mathbf{360^\circ}}\pi\left(\mathbf{r_2^2} - \mathbf{r_1^2}\right) = \frac{\theta}{\mathbf{360^\circ}}\pi(\mathbf{r_2} + \mathbf{r_1})(\mathbf{r_2} - \mathbf{r_1})$$

Worked Archetype 5 | Standard Board Exam Horse Grazing Model

Problem: A horse is tied to a peg at one corner of a square grass field of side 15 m by a 5 m long rope. Find: (i) the area in which the horse can graze, and (ii) the increase in grazing area if the rope were 10 m long. (Use $\pi = 3.14$)

(i) Initial grazing area ($r_1 = 5$ m, $\theta = 90^\circ$):

$$A_1 = \frac{90^\circ}{360^\circ} \times \pi r_1^2 = \frac{1}{4} \times 3.14 \times 25 = \mathbf{19.625}\ \text{m}^\mathbf{2}$$

(ii) Grazing area with extended rope ($r_2 = 10$ m):

$$A_2 = \frac{1}{4} \times \pi r_2^2 = \frac{1}{4} \times 3.14 \times 100 = 78.5\ \text{m}^2$$

Increase in grazing area (ΔA):

$$\mathbf{\Delta A} = A_2 - A_1 = 78.5 - 19.625 = \mathbf{58.875}\ \text{m}^\mathbf{2}$$

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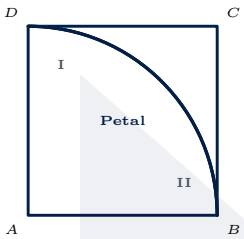
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Module 8 | Composite Overlaps: The Famous Petal / Leaf Geometry

8.1 The Intersecting Quadrants "Leaf / Lens" in a Square

When two circular quadrants of radius a are drawn inside a square of side a with opposite vertices as centers, they overlap to form a central **leaf/petal shaped region**:



Master Algebraic Derivation of the Petal Area:

1. $Q_1 + Q_2 =$ Area of Square + Area of Petal.
2. $\mathbf{A_{petal} = 2\left(\frac{1}{4}\pi a^2\right) - a^2 = \left(\frac{\pi}{2} - 1\right)a^2}$.
3. For $\pi = \frac{22}{7}$:

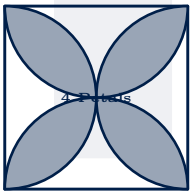
$$\mathbf{A_{petal} = \left(\frac{11}{7} - 1\right)a^2 = \frac{4}{7}a^2}$$

4. **Unshaded Corners (I + II):**

$$\mathbf{A_{unshaded} = a^2 - \frac{4}{7}a^2 = \frac{3}{7}a^2}$$

8.2 The Four-Petal Flower in a Square (Overlapping Semicircles)

When four semicircles are drawn on the four sides of a square as diameters, they intersect to create a **four-petaled symmetrical flower**:



Four-Petal Flower Equations (Side a, $r = a/2$):

- Total Area of 4 Semicircles:

$$4 \times \left(\frac{1}{2}\pi r^2\right) = 2\pi(a/2)^2 = \frac{\pi a^2}{2}$$

- Total Area of 4 Petals:

$$\mathbf{A_{4\ petals} = \frac{\pi a^2}{2} - a^2 = \left(\frac{\pi}{2} - 1\right)a^2 = \frac{4}{7}a^2}$$

- **Identity:** The single large leaf and the 4 small petals have the **exact same area** $\mathbf{\frac{4}{7}a^2}$!

Worked Archetype 6 | Designed Petal Region in a Square

Problem: Calculate the area of the designed region common between two quadrants of circles of radius 8 cm each inside a square of side 8 cm. (Take $\pi = \frac{22}{7}$)

Solution: Here $a = 8$ cm. Using the direct Maths Manthan Petal Formula:

$$\text{Area} = \left(\frac{\pi}{2} - 1\right)a^2 = \frac{4}{7}a^2 = \frac{4}{7} \times 8^2 = \frac{256}{7} = \mathbf{36.57}\ \text{cm}^2$$

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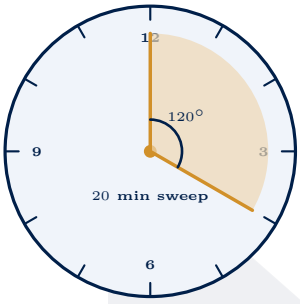
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Module 9 | Kinematic Sweeps: Clock Hands & Wiper Blades

9.1 The Angular Velocities of Clock Hands

A circular clock dial is divided into 12 hours and 60 minutes (360° circle):



Angular Velocities of Clock Hands:

- **Minute Hand:**

$$\text{Speed} = \frac{360^\circ}{60 \text{ min}} = \mathbf{6}^\circ/\text{minute}$$

$$\text{Angle in } \mathbf{m} \text{ minutes: } \theta = \mathbf{6m}^\circ$$

- **Hour Hand:**

$$\text{Speed} = \frac{360^\circ}{720 \text{ min}} = \mathbf{0.5}^\circ/\text{minute} = \left(\frac{\mathbf{1}}{\mathbf{2}}\right)^\circ/\text{min}$$

$$\text{In 1 hour (60 min): } \theta = \mathbf{30}^\circ$$

9.2 Automobile Windshield Wipers (Twin Sweeps)

$$\text{Total Area Cleaned at Each Sweep} = \mathbf{2} \times \left(\frac{\theta}{\mathbf{360^\circ}} \times \pi \mathbf{r^2}\right)$$

Worked Archetype 7 | Minute Hand Swept Area

Problem: The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes. (Take $\pi = \frac{22}{7}$)

Step 1: Angle swept: $\theta = 5 \text{ min} \times 6^\circ/\text{min} = 30^\circ$.

Step 2: Area swept: $\text{Area} = \frac{30^\circ}{360^\circ} \times \frac{22}{7} \times 14^2 = \frac{1}{12} \times 22 \times 2 \times 14 = \mathbf{\frac{154}{3}}$ **cm²** $\approx$ **51.33** **cm²**.

Worked Archetype 8 | Car Windshield Twin Wipers

Problem: A car has two non-overlapping wipers with blade length 25 cm sweeping angle 115°. Find total cleaned area per sweep. (Take $\pi = \frac{22}{7}$)

$$\text{Total Area} = 2 \times \left(\frac{115^\circ}{360^\circ} \times \frac{22}{7} \times 25^2\right) = 2 \times \frac{23}{72} \times \frac{22}{7} \times 625 = \mathbf{\frac{158125}{126}} \text{ cm}^\mathbf{2} \approx \mathbf{1254.96} \text{ cm}^\mathbf{2}$$

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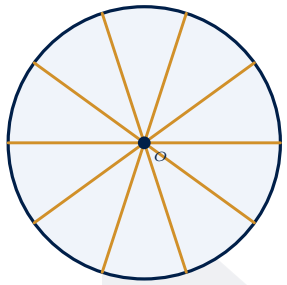
CORE SYLLABUS

AREAS RELATED TO CIRCLES

Module 10 | Advanced Combinations: Brooches & Hippocrates Lunulae

10.1 The Circular Brooch with Silver Wire Diameters

A popular board exam configuration involves a decorative circular brooch crafted with silver wire:



10 Equal Sectors
$\theta = \frac{360^\circ}{10} = 36^\circ$

Equations for Brooch of Diameter d with k Diameters:

- **Total Length of Silver Wire (L):**

$$\mathbf{L = \pi d + k \cdot d = d(\pi + k)}$$

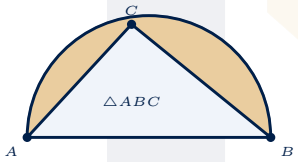
For standard board problem ($k = 5$ diameters):

$$\mathbf{L = \pi d + 5d = d\left(\frac{22}{7} + 5\right) = \frac{57}{7}d}$$

- **Central Angle of Each Sector:** $\theta = \frac{360^\circ}{2k} = 36^\circ$
- **Area of Each Sector:** $\mathbf{A = \frac{1}{10}\pi r^2 = \frac{\pi d^2}{40}}$

10.2 Hippocrates Lunulae (The Crescent Moon Quadrature Theorem)

Curved areas can equal rectilinear areas:



Theorem of Hippocrates (Sum of Lunulae): In a right-angled triangle $\triangle ABC$ ($\angle C = 90^\circ$):

1. Semicircles are drawn on legs a, b and hypotenuse c.
2. **Theorem:** The sum of the areas of the two shaded crescents (lunulae) equals the area of $\triangle ABC$!

$$\text{Area}(\text{Lune}_1) + \text{Area}(\text{Lune}_2) = \text{Area}(\triangle \mathbf{ABC}) = \frac{1}{2}\mathbf{ab}$$

Worked Archetype 9 | Brooch Silver Wire & Sector Calculation

Problem: A brooch is made with silver wire as a circle with diameter 35 mm and 5 diameters dividing it into 10 equal sectors. Find: (i) wire length, and (ii) area of each sector.

(i) Total Wire Length (L): $L = \pi d + 5d = \left(\frac{22}{7} \times 35\right) + (5 \times 35) = 110 + 175 = \mathbf{285}$ mm.

(ii) Area of each sector (A): $A = \frac{1}{10}\pi r^2 = \frac{1}{10} \times \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = \frac{\mathbf{385}}{\mathbf{4}}$ mm^2 $= \mathbf{96.25}$ mm^2.

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QUICK REFERENCE

AREAS RELATED TO CIRCLES

Module 11 | Master Formula Matrix & Decision Protocol

11.1 The Comprehensive Mensuration Matrix for Circular Geometries

Geometric Figure	Parameters	Boundary / Perimeter (P)	Enclosed Surface Area (A)
Full Circle	$r, d = 2r$	$C = 2\pi r = \pi d$	$A = \pi r^2 = \frac{1}{4}\pi d^2$
Semicircle	r	$P = \pi r + 2r = r(\pi + 2)$	$A = \frac{1}{2}\pi r^2$
Quadrant	r	$P = \frac{1}{2}\pi r + 2r = r\left(\frac{\pi}{2} + 2\right)$	$A = \frac{1}{4}\pi r^2$
Circular Sector	r, θ	$P = \frac{\theta}{360^\circ}(2\pi r) + 2r$	$A = \frac{\theta}{360^\circ}\pi r^2 = \frac{1}{2}lr$
Minor Segment	r, θ	$P = \text{Arc } l + 2r\sin(\frac{\theta}{2})$	$A = \frac{\theta}{360^\circ}\pi r^2 - \frac{1}{2}r^2\sin\theta$
Major Segment	r, θ	$P = (2\pi r - l) + 2r\sin(\frac{\theta}{2})$	$A = \pi r^2 - A_{\text{minor segment}}$
Circular Ring (Annulus)	R, r	$P_{\text{total}} = 2\pi(R + r)$	$A = \pi(R^2 - r^2) = \pi(R + r)(R - r)$
Square Leaf / Petal	a	$P = 2 \times \left(\frac{1}{2}\pi a\right) = \pi a$	$A = \left(\frac{\pi}{2} - 1\right)a^2 \approx \frac{4}{7}a^2$

11.2 The 3-Step Board Examination Decision Flowchart

STEP 1: Boundaries
Curved only, or curved + flat?
($P_{\text{sector}} = l + 2r$)

→

STEP 2: Classify Region
Sector, Segment, or Annulus?
($A_{\text{segment}} = \text{Sector} - \triangle$)

→

STEP 3: Evaluate $\triangle OAB$
$60^\circ \implies \frac{\sqrt{3}}{4}r^2$, $90^\circ \implies \frac{1}{2}r^2$
Factor before evaluating!

Fast Arithmetic Factoring Constants (When $\pi = \frac{22}{7}$):

- **Quadrant Perimeter:** $P = \frac{1}{2}\pi r + 2r = \left(\frac{11}{7} + 2\right)r = \mathbf{\frac{25}{7}r}$
- **Semicircle Perimeter:** $P = \pi r + 2r = \left(\frac{22}{7} + 2\right)r = \mathbf{\frac{36}{7}r}$
- **Shaded Corners of Square ($a = 2r$):** Area $= 4r^2 - \pi r^2 = \left(4 - \frac{22}{7}\right)r^2 = \mathbf{\frac{6}{7}r^2} = \mathbf{\frac{3}{14}a^2}$
- **Single Petal in Square:** Area $= \left(\frac{\pi}{2} - 1\right)a^2 = \left(\frac{11}{7} - 1\right)a^2 = \mathbf{\frac{4}{7}a^2}$

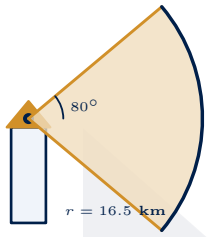
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AREAS RELATED TO CIRCLES

Module 12 | Board Exam Case-Study & Applied Modeling

Case Study 1 | Navigation Marine Lighthouse Warning Sector

To warn ships of underwater rocks, a coastal lighthouse spreads a red colored light over a sector of angle 80° to a distance of 16.5 km.



4-Mark Board Question Protocol

Questions:

1. What geometric figure represents the illuminated sea surface? [1M]
2. Calculate the length of the outer warning arc on the ocean. [1M]
3. Find the area of the sea over which ships are warned. ($\pi = 3.14$) [2M]

Solution & Marking Scheme Evaluation

1. **Geometric Classification [1M]:** The illuminated region is a **sector of a circle** with radius $r = 16.5$ km and central angle $\theta = 80°$.
2. **Length of Warning Arc [1M]:** $l = \frac{80°}{360°} \times 2 \times 3.14 \times 16.5 = \frac{2}{9} \times 103.62 = \mathbf{23.03}$ km.
3. **Total Area of Sea Warned [2M]:** $A = \frac{80°}{360°} \times 3.14 \times (16.5)^2 = \frac{2}{9} \times 3.14 \times 272.25 = \mathbf{189.97}$ km$^\mathbf{2}$.

Case Study 2 | Architectural Roundabout with Floral Sectors

A city municipal corporation designs a circular traffic roundabout of radius 28 m. At the center is a circular fountain of radius 7 m. The remaining annular region is divided into 6 congruent sectors: 3 sectors planted with red roses, and 3 alternating sectors with green grass.

Case Study Questions & Structured Solutions

1. **Find the central angle of each sector:** $\theta = \frac{360°}{6} = \mathbf{60°}$.
2. **Find the area of the central fountain:** $A_{\text{fountain}} = \pi r^2 = \frac{22}{7} \times 7^2 = \mathbf{154}$ m$^\mathbf{2}$.
3. **Find the total area planted with red roses:**

$$\text{Total Annular Area} = \pi(R^2 - r^2) = \frac{22}{7}(28^2 - 7^2) = \frac{22}{7}(784 - 49) = \frac{22}{7} \times 735 = 2310 \text{ m}^2$$

Since roses occupy 3 out of 6 equal sectors: Area of Roses $= \frac{1}{2} \times 2310 = \mathbf{1155}$ m$^\mathbf{2}$.

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EXAM STRATEGY

AREAS RELATED TO CIRCLES

Module 13 | Board Traps, Verification Hacks & 5-Point Audit

13.1 The Top 7 Board Examination Blunders (And How to Neutralize Them)

Trap 1: The Sector Perimeter Omission: Writing $P = \frac{\theta}{360^\circ}(2\pi r)$ and forgetting to add the two radii.

TRAP ALERT

Always write $\mathbf{P = l + 2r}$ before substituting numbers!

Trap 2: Diameter Substituted as Radius: When given "diameter 14 cm", blindly writing $r = 14$. Always write $\mathbf{r = d/2 = 7}$ cm in the first line of your solution.

Trap 3: Wrong π Value Substituted: Using $\frac{22}{7}$ when instructed "(Use $\pi = 3.14$)". This incurs a direct deduction under CBSE marking schemes!

Trap 4: Segment Addition Blunder: Adding the triangle to the sector instead of subtracting: Segment = Sector $- \triangle OAB$.

Trap 5: Equilateral vs Obtuse Triangle Confusion: Using $\frac{1}{2}r^2$ for $\theta = 60^\circ$ instead of $\frac{\sqrt{3}}{4}\mathbf{r^2}$.

Trap 6: Clock Angle Calculation Error: Assuming the minute hand moves 1° per minute. It moves $\mathbf{6}^\circ$/minute! The hour hand moves $\mathbf{0.5}^\circ$/minute.

Trap 7: Mixed Linear Units: Calculating area using radius in centimeters and length in meters. Convert all dimensions to the same unit (cm or m) before calculating!

13.2 High-Speed Algebraic Factorization Protocol

The Slow Method (High Arithmetic Risk)	The Maths Manthan Factoring Method
Find annulus area ($R = 14, r = 7$): Area $= \frac{22}{7} \times 14^2 - \frac{22}{7} \times 7^2$ $= \frac{22}{7} \times 196 - \frac{22}{7} \times 49$ $= 616 - 154 = \mathbf{462}$ cm$^\mathbf{2}$	Factor algebraic terms first: Area $= \pi(R^2 - r^2) = \pi(\mathbf{R + r})(\mathbf{R - r})$ Substitute once at the end: $= \frac{22}{7} \times (14 + 7) \times (14 - 7) = \frac{22}{7} \times 21 \times 7 = \mathbf{462}$ cm$^\mathbf{2}$

13.3 The 5-Point Board Examination Pre-Submission Checklist

Final 60-Second Answer Sheet Verification Protocol:

- ☐ **Dimension Check:** Did you state $r = d/2$ explicitly with correct unified units (cm or m)?
- ☐ **Symbolic Formula Stated:** Did you write the algebraic formula before substituting numbers? (Guarantees CBSE method marks).
- ☐ **Specified π Respected:** Did you check whether $\pi = \frac{22}{7}$ or $\pi = 3.14$ was requested?
- ☐ **Perimeter vs Arc Length:** If asked for perimeter of a sector, did you remember to add $+2r$?
- ☐ **Dimensional Units Attached:** Lengths in cm/m, Areas in cm$^\mathbf{2}$/m$^\mathbf{2}$, Angles in °!

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